Alexander Urs-Badet

Full-Stack Software Engineer | Angular · AWS · Node.js

Lynnwood, WA | (425) 678-5591 | alexanderursbadet@gmail.com | LinkedIn · Portfolio

SUMMARY

Full-stack software engineer building Angular/AWS systems for federally regulated environments serving 1,500+ users: production forms, financial calculators, and legacy modernization with high Jest coverage. Independent systems projects in real-time computer vision, algorithmic trading, and network reverse-engineering. Focused on well-tested software that holds up in production, end to end.

TECHNICAL SKILLS

Frontend	Angular, React, TypeScript, JavaScript, HTML/CSS, Tailwind CSS
Backend & Cloud	Node.js, Python, Java, Spring Boot, C#, AWS (Lambda, S3, API Gateway), REST APIs, SQL
Testing & Delivery	Jest, unit testing, CI/CD, Git, Agile/Scrum, code review, legacy migration
Systems	WebSockets, asyncio, SQLite, network protocols, reverse engineering
AI / ML	PyTorch, TensorFlow, scikit-learn, OpenCV, DeepLabCut
Specializations	Form-heavy Angular apps, financial calculators, legacy migration (Struts to Angular/Spring Boot), real-time systems

PROFESSIONAL EXPERIENCE

Software Engineer | TSPi | Lynnwood, WA · 11/2023 – Present

- Led full-stack development of production systems serving 1,500+ FSA loan officers and applicants across the U.S. Farm Service Agency.
- Architected and deployed a financial optimization calculator (AWS Lambda, Node.js, Angular) that reduced manual calculation overhead by ~40%.
- Modernized legacy Java/Struts workflows into Angular/Spring Boot features while maintaining extensive Jest coverage across critical front-end components.
- Developed features for a nationwide debt-consolidation system presented to senior FSA and congressional stakeholders.

Software Engineering Intern | T-Mobile | Bellevue, WA · 07/2021 – 08/2021, 07/2022 – 08/2022

- Optimized core sorting and data processing algorithms used in Sync-Up product; achieved 50% runtime reduction.
- Built Python/Flask applications integrating external APIs; consolidated multiple fragmented data trackers into unified application.
- Mentored 2 junior interns in Python; enabled both to successfully deliver 2 projects independently, raising internal engineering capacity.

PROJECTS

Reverse-Engineered Game Server & Combat Simulator · Python, asyncio, Protobuf, FastAPI

- Reconstructed a live mobile game's TCP/protobuf wire protocol from captured traffic and a decompiled client; built a ~14k-line asyncio server (TCP + HTTP asset patcher) that drives the real client from boot to home screen.
- Shipped a FastAPI web app around it: roster viewer, team builder, and a fight simulator with a support-slot optimizer, backed by a combat engine calibrated against captured fights.

Algorithmic Trading Systems · Python, Options, IBKR API, WebSockets

- Built an intraday options system with pluggable signal engines (VWAP-pullback, opening-range breakout, mean-reversion), a multi-layer risk engine (position sizing, loss lockouts, kill switch), and a walk-forward backtester.
- Rebuilt it risk-first as a paper-only IBKR bot: fail-closed entry gates, durable loss latches, native combo orders, a real-time dashboard, and a reproducible evidence bundle (API/WebSocket probes, SQLite state, restart comparison).

Real-Time Tracking & Robot Control · Python, OpenCV, Kalman Filters, Robotics

- Built an OpenCV detection pipeline (LAB/HSV color segmentation, morphology, contour centroids) on a threaded 30 FPS capture loop that never tracks a stale frame.
- Wrote per-object constant-velocity Kalman filters with gated association and occlusion handling, then closed the loop onto a mobile robot over a hardware interface.

EDUCATION

- A.A. & Full Stack Developer Certificate · Edmonds College, Edmonds, WA · 2022–2023
- B.S. Psychology Coursework · University of Washington, Seattle